

SECTION B

Answer all questions in the spaces provided.

7. (a) Ammonia is an example of a weak base. Describe what is meant by the term *weak base*. [1]

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- (b) A mixture of ammonia and ammonium chloride in aqueous solution can be used as a basic buffer solution. Explain what is meant by a *buffer solution* and how this mixture can act as a buffer solution. [3]

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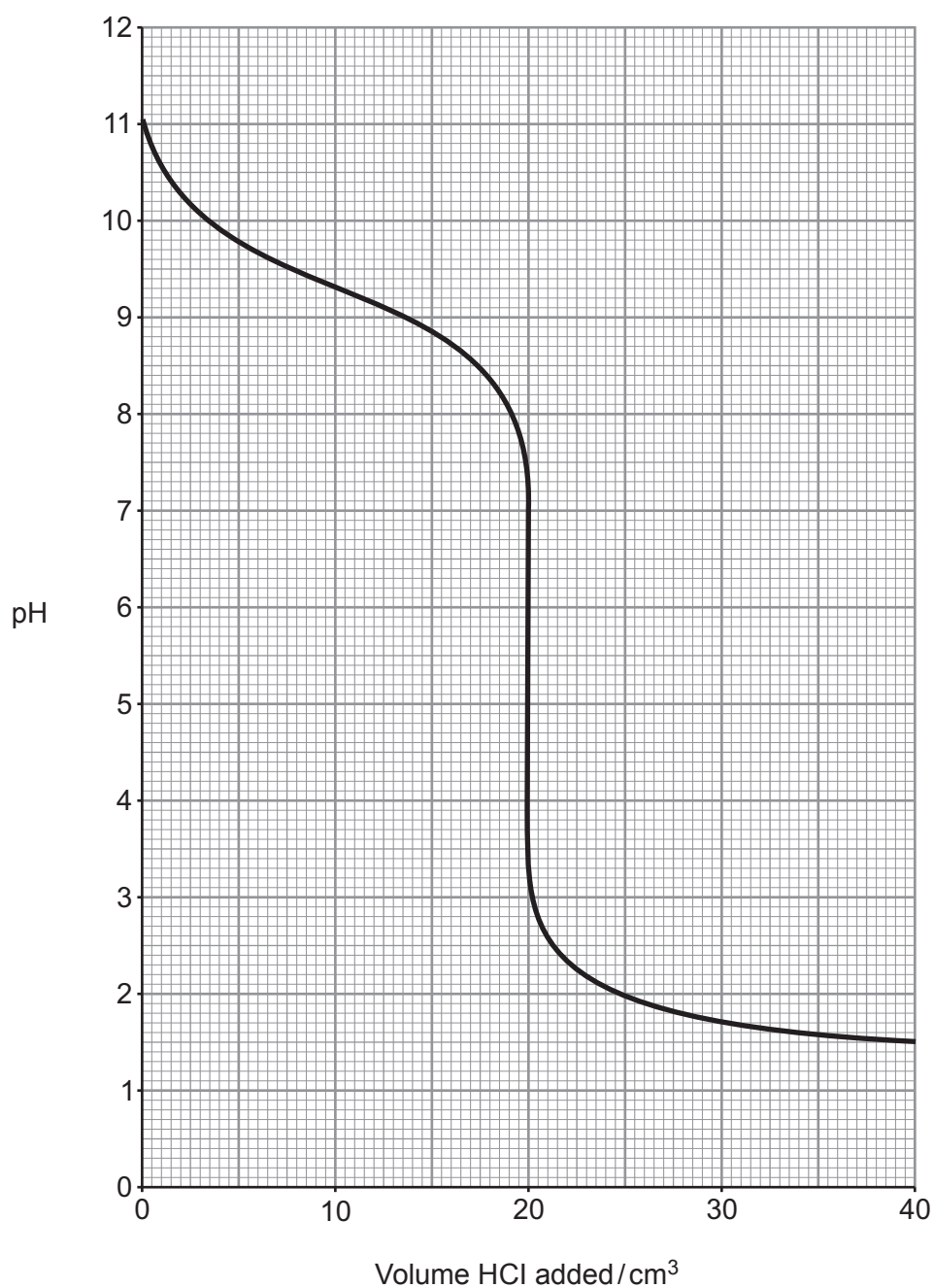
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- (c) The pH curve for titration of a solution of ammonia against hydrochloric acid of an equal concentration is given below.



- (i) Calculate K_a for the ammonium ion.

[2]

$K_a = \dots\dots\dots \text{mol dm}^{-3}$



- (ii) Select an appropriate indicator for this titration from the list below, giving a reason for your answer. [1]

Indicator	pH range
bromothymol blue	6.0 – 7.6
4-nitrophenol	5.0 – 7.0
methyl yellow	2.9 – 4.0
phenolphthalein	8.2 – 10.0

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